

Darin Tsui

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Education

Georgia Institute of Technology, Ph.D. in Electrical and Computer Engineering Aug 2023 – May 2028
• Advisor: Amirali Aghazadeh

University of California San Diego, B.S. in Bioengineering Sep 2019 – Jun 2023

Selected Publications

Darin Tsui, Kunal Talreja, Daniel Saeedi, and Amirali Aghazadeh. Protein circuit tracing via cross-layer transcoders. *arXiv preprint arXiv:2602.12026*, 2026.

Zain Shabeeb*, Daniel Saeedi*, **Darin Tsui***, Vida Jamali, and Amirali Aghazadeh. cryoSENSE: Compressive sensing enables high-throughput microscopy with sparse and generative priors on the protein cryo-EM image manifold. In *Conference on Computer Vision and Pattern Recognition (CVPR) (Accepted)*, 2026.

Darin Tsui, Aryan Musharaf, Justin S. Kang, Yigit E. Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

Honors & Awards

Steve W. Chaddick Fellowship, Georgia Tech 2025
• Fellowship awarded for outstanding contributions in research.

NSF Graduate Research Fellowship 2024 – 2027
• Fellowship recognizing potential for significant research contributions (top 14% of applicants).

Experience

Graduate Research Assistant, AI ML and Information Research Group – Atlanta, GA Jun 2023 – Present
• Leveraged *mechanistic interpretability* frameworks, such as cross-layer transcoders and sparse autoencoders, for circuit discovery, protein design, and extracting biological mechanisms from protein language models.
• Extracted higher-order interactions in models 1000-fold quicker than state-of-the-art methods by developing algorithms at the intersection of *signal processing*, *coding theory*, *combinatorics*, and *interpretability*.
• Developed principled *generative models* for high-throughput cryo-electron microscopy (cryo-EM), autoregressive protein generation, and variant effect prediction.

Data Engineering Intern, Surgalign – San Diego, CA Jan 2023 – Aug 2023
• Interned at Surgalign, a small medical device company focused on developing AI solutions in spinal surgery (now acquired by Xtant Medical and Augmetics).
• Designed an internal MRI image processing application using Python that reduced preprocessing time.
• Assisted in developing ground truth dataset of imaging data and validated deep learning models against surgeon-generated data using Sørensen-Dice coefficient statistical testing.

Leadership

Reviewer: 2026 ICML, ECCV, CVPR, ICLR GEM; 2025 NeurIPS AI4Science; 2024 ICLR GEM.

Capstone Project Advisor, UC San Diego – San Diego, CA Jul 2023 - Jun 2024
• Mentored two undergraduates toward the design and development of a surgical navigation system with 3D tracking and augmented reality.
• Capstone project team won the *Teaching Assistant's Choice Award* at Bioengineering Day.